HEAPLINE · PROJECT PLAN

HeapLine project plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heapline		
Source	docs/11-plan.md @ 0055524		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-03	Amish Chadha	First plan, written for the kit 1.9.1 upgrade

HeapLine project plan

The project is at stage P3 (TRL 3, paper proof of concept) with the pipeline documents and the D3 model of every concept option in place. The next gate is Amish's review of the concept options and of the decisions listed in `docs/REVIEW.md`; no TRL 4 work starts until he asks. Method: STANDARDS sections 19 and 23.

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Problem and users	Project lead	HPL-PRB-001 with users and stakeholders	Done (P3)
STEM check	Design lead	HPL-STM-001, `stem_check.py`	Done (P3)
Concept options	Design lead	HPL-OPT-001, `options_calc.py`, three D3 models	Done; choice awaiting Amish
Requirements	Design lead	HPL-URS-001, HPL-PRS-001, HPL-RTM-001	Done (P3)
CAD and drawings	Design lead	`cad/src/model.py`, STEP, STL, HPL-DWG-001 and 002	Done (P3, D3)
Calculations	Design lead	HPL-CAL-001	Done (P3)
Build plan	Design lead	HPL-BLD-001, HPL-DWG-101 to 116	Done (P3, plan only)
Regulatory, feasibility, risk, market	Project lead	HPL-REG-001, FEA, RSK, MKT	Done (P3)
Co-design partner	Project lead	Agreement with a cooperative and a Red Cross branch or fire service	Not started
Prototype build and tests	Fabricator	Build log, release and pull tests, probe trials, drills	Planned, TRL 4 (on hold while the stage cap is P3)

Table 2. Resources.

Resource	Deliverable	Effort	Estimated cost (USD)
Co-design partner	Site trials and drills	To assess	To assess
Fabricator with a small welder	First prototype of the capstan and the search kit	To assess	About 1,374 in parts for the host site (HPL-CAL-001 G3), indicative, not a quote
Test site (a waste heap)	Probe trials and board load tests	To assess	To assess
Reviewer for regulatory questions	HPL-REG-001 items marked "to assess"	To assess	To assess

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P3	Pipeline documents, D3 model of every option, constructable design, build plan, priced BOM	Reached
P4	Co-design partner agreed; prototype built and tested	On hold (stage cap)
P5	Pilot at one site with drills	On hold (stage cap)
P6	Release and handover	On hold (stage cap)

Next steps and decision requested

1. Amish chooses among the concept options (HPL-DDR-005, "Proposed, awaiting Amish").
2. Amish decides whether to approach the co-design partner and start the TRL 4 plan; none of it begins before he asks.

References

Every source cited in this repo's documents is listed in its own Sources section (HPL-PRB-001, HPL-STM-001, HPL-REG-001, HPL-MKT-001) and in the Chicago-style list in `docs/REVIEW.md`.